

# TEJAS PANDYA

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## EDUCATION

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| <b>NEW YORK UNIVERSITY, TANDON SCHOOL OF ENGINEERING</b> , Brooklyn, NY                              | Expected 05/27 |
| Master of Science in Financial Engineering   GPA: 3.8/4.0                                            |                |
| Graduate Teaching Assistant: Financial Economics; C++ Financial Software Laboratory; Value Investing |                |
| <b>SVKM'S NMIMS, SCHOOL OF BUSINESS MANAGEMENT</b> , Mumbai, India                                   | 04/22          |
| Master of Business Administration, Concentration in Finance   GPA: 3.3/4.0                           |                |
| <b>VELLORE INSTITUTE OF TECHNOLOGY</b> , Vellore, India                                              | 06/20          |
| Bachelor of Technology in Mechanical Engineering   GPA: 8.9/10.0                                     |                |

## PROGRAMMING / TECHNICAL SKILLS / CERTIFICATIONS

**Programming:** C/C++ (C++20, STL, Boost), Python, SQL, Bash, LaTeX/Typst  
**Libraries & Frameworks:** NumPy, Pandas, Polars, SciPy, scikit-learn, statsmodels, LightGBM, XGBoost, PyTorch, pytest  
**Data & Tools:** Linux/Unix, Git, CMake, Streamlit, Shiny, Excel, Bloomberg Terminal, Tableau, Power BI  
**Certifications:** CFA Level I; C++ Programming for Financial Engineering (Baruch College); Bloomberg Market Concepts; CFI FMVA

## COURSEWORK HIGHLIGHTS

**Quantitative Methods & Computing:** Financial Econometrics, Machine Learning in Financial Engineering, Quantitative Methods in Finance, Financial Computing, Applied Numerical Methods, Data Structures and Algorithms  
**Risk, Valuation & Derivatives:** Financial Risk Management, Valuation for Financial Engineering, Quantitative Portfolio Management, Options Pricing & Stochastic Calculus, Fixed Income Securities and Interest Rate Derivatives

## EXPERIENCE

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| <b>NEW YORK UNIVERSITY, TANDON SCHOOL OF ENGINEERING</b> , Brooklyn, NY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 05/26 - 08/26 |
| <i>Quantitative Researcher, QWIM Industry Capstone</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |               |
| <ul style="list-style-type: none"><li>Implemented a log-periodic power-law singularity bubble-confidence tilt as the fourth ablation level above regime-conditioned covariance blending and quantile-regression tail estimation, the strongest of eight variants, holding Sharpe 0.56 against 0.50 on the 2008-2025 window</li><li>Diagnosed thirteen environment-configuration defects behind a 278-file team pytest suite that reported success while executing zero tests on macOS, restoring 10,423 tests to green and documenting the 22 pre-existing failures rather than leaving them quietly reported as green</li><li>Built the Regime Risk Budgeting dashboard tab and its Typst client-report exporter, 1,058 lines of Python across ten modules with twelve unit tests, written fail-safe so that any missing artifact degrades only that one report section rather than breaking the whole client report</li></ul>                                                                                                                                                                                                                                                                                                                      |               |
| <b>NOMURA SERVICES INDIA PRIVATE LIMITED</b> , Mumbai, India                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 08/22 - 07/25 |
| <i>Senior Risk Analyst, Market Risk Infrastructure</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |               |
| <ul style="list-style-type: none"><li>Owned the lifecycle of VaR and sensitivity models across equities, rates, FX and commodities, from daily production runs through backtesting; Python anomaly detection triaged exceptions to stale marks, erroneous data or booking errors, cutting investigation turnaround 40% and restoring SLAs</li><li>Delivered quarterly Basel 2.5 and FRTB stress-testing and capital submissions alongside front-office and regulatory teams, engineering rules-based validation over delta, vega and curvature sensitivities that reconciled fully against front-office pricing and caught anomalies pre-submission</li><li>Engineered Python and SQL pipelines across equities, rates, FX and commodities that aggregate, validate and reconcile daily P&amp;L, positions and Level-1/Level-2 market data into the VaR and sensitivity dashboards, eliminating more than 50 hours of manual processing every month</li><li>Led a three-person workstream on US-region end-of-day risk reporting, cutting overnight batch runtime 35% through SQL query optimisation and job scheduling, and built the VaR, P&amp;L and exception dashboards that senior stakeholders read each morning before the US open</li></ul> |               |

## ACADEMIC PROJECTS

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| <b>TREASURY-CURVE SWAP VALUATION &amp; HISTORICAL COUNTERPARTY RISK</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 05/26 |
| <i>Python, NumPy, SciPy</i>   <a href="https://github.com/tejaspandya9598/sofr-swap-pnl-attribution">github.com/tejaspandya9598/sofr-swap-pnl-attribution</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |       |
| <ul style="list-style-type: none"><li>Priced historical counterparty CVA from 20,000 paired exposure paths over 8,478 days of joint rate and credit history, measuring wrong-way risk at \$89,553, 16.9% above a decoupled baseline sharing its curve paths bit-for-bit, at a \$2.35M peak expected exposure, with no collateral agreement</li><li>Simulated curves inside PCA factor space, three components covering 98% of curve variance, after unconstrained pillar walks reached 740bp of 2s30s against the 401bp maximum the market has printed and produced curves carrying no positive discount factor at the long end</li></ul>   |       |
| <b>HFT LIMIT ORDER BOOK &amp; MATCHING ENGINE</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 02/26 |
| <i>C++20, CMake, CTest</i>   <a href="https://github.com/tejaspandya9598/hft-matching-engine">github.com/tejaspandya9598/hft-matching-engine</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |
| <ul style="list-style-type: none"><li>Built a deterministic price-time-priority matching engine timing roughly 21M messages per second on an Apple M2 across five million crossing synthetic limit orders, and 10.7M through the lock-free end-to-end pipeline, on a zero-allocation hot path with 64-byte cache-aligned structures</li><li>Reached O(1) average add, cancel and execute through intrusive per-level lists with hash lookup, a new price level costing O(log L) for best-bid tracking; 20 CTest assertions plus five benchmark checks pin price-time priority, partial fills and pool reuse under 10k-order churn</li></ul> |       |
| <b>OPTIONS PRICING ENGINE - VANILLA, AMERICAN AND BARRIER</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 10/25 |
| <i>C++20, CMake, CTest</i>   <a href="https://github.com/tejaspandya9598/options-pricing-engine">github.com/tejaspandya9598/options-pricing-engine</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |       |
| <ul style="list-style-type: none"><li>Implemented three independent valuation methods behind one interface: closed-form Black-Scholes with analytic greeks, Monte Carlo with exact terminal sampling, and a Crank-Nicolson solver using projected SOR for American early exercise, with barriers as domain boundaries</li><li>Cross-validated Black-Scholes, Monte Carlo and Crank-Nicolson to within 0.6% on an at-the-money European call, pinning that agreement as regression tests at \$0.02 and \$0.06 tolerances, recovering a 0.51 early-exercise premium and documenting the discrete-monitoring gap on barriers</li></ul>         |       |